

Question 1 : What is Power BI and why is it used in businesses?

Power BI is a business intelligence (BI) tool by Microsoft that helps organizations visualize, analyze, and share data.

It connects to multiple data sources, transforms raw data into interactive dashboards and reports, and provides insights in a simple, visual way.

Use Power BI :-

1. **Data Visualization** - Turns complex data into easy-to-understand charts, graphs, and dashboards.
2. **Better Decision-Making** - Helps managers and executives make informed decisions using real-time data insights.
3. **Integration of Multiple Data Sources** - Combines data from databases, Excel, cloud apps, and other systems into one unified view.
4. **Real-Time Analytics** - Allows businesses to monitor KPIs and metrics instantly.
5. **Improves Efficiency** - Automates reports and reduces time spent manually analyzing data.

Question 2 : Name and explain the three main components of Power BI.

- **Power BI Desktop:** A Windows desktop application used by data analysts to connect to, transform, and model data from various sources. It allows users to create reports with interactive visuals, incorporating DAX (Data Analysis Expressions) for complex calculations.
- **Power BI Service (SaaS):** A cloud-based platform used to publish reports, create dashboards, schedule data refreshes, and share insights securely across an organization. It enables collaboration and central management of datasets.
- **Power BI Mobile Apps:** A mobile application available on iOS, Android, and Windows devices that allows users to access, interact with, and monitor their dashboards and reports securely on the go.

Question 3 : Explain the Power BI workflow.

It has four main steps:

- **Get Data** - Connect to different data sources like Excel, SQL databases, cloud apps, or websites. Collect all the raw data you need for analysis.
- **Transform & Clean Data** - Use Power Query to remove errors and duplicates, fix missing values, standardize formats. This step makes the data accurate and ready for reporting.

- **Build Reports & Visuals** - Use Power BI Desktop to create Charts, tables, maps, KPIs, and dashboards . Combine visuals to make insights easy to understand.
- **Publish & Share** - Publish reports to Power BI Service (cloud). Share dashboards with teams or managers. Users can also access reports on mobile devices.

Question 4 : List any four data cleaning tasks that can be performed in Power Query

Here are four common tasks:

- **Remove Duplicates** - Deletes repeated rows to ensure each record is unique.
- **Handle Missing Values** - Fill missing values, replace them with defaults, or remove incomplete rows.
- **Change Data Types** - Ensure columns have the correct type, like number, text, or date.
- **Trim & Clean Text** - Remove extra spaces, unwanted characters, or fix inconsistent text formatting.

Question 5 : Write step by step instructions to load the above dataset.

Step 1: Open Power BI Desktop - Launch Power BI Desktop on your computer.

Step 2: Get Data - Click Home → Get Data → Excel / CSV / Database (depending on your dataset file). Browse and select your dataset file. Click Load .

Step 3: Transform Data (Optional) - Click Transform Data to open Power Query Editor.

- Perform data cleaning tasks:
 - ✓ Remove duplicates
 - ✓ Fix missing values
 - ✓ Correct data types (e.g., Sales_Amount = number, Txn_Date = date)
 - ✓ Standardize text fields (e.g., Customer Name)
- Click Close & Apply to save changes.

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

a) Data View

- Shows a table-like view of your data after it has been loaded.
- Purpose:
 - Inspect the data to ensure it is clean and correct
 - Check column values, data types, and transformations

- Example Use: Verify Sales_Amount values or Customer_ID after cleaning in Power Query.

b) Report View

- The visualization canvas where you create charts, graphs, tables, and dashboards.
- Purpose:
 - Build interactive reports
 - Drag fields to create visuals and KPIs
 - Arrange multiple visuals on a single page
- Example Use: Create a bar chart showing total sales by region or a line chart for daily transactions.

c) Model View

- Shows the relationship diagram between tables in your dataset.
- Purpose:
 - Define relationships between tables (like Sales and Customers)
 - Ensure proper data connections for accurate reporting
 - Manage primary keys, foreign keys, and cardinality
- Example Use: Link Customer_ID in the Sales table to CustomerID in the Customers table to avoid duplicate or disconnected data.

Question 7 : Discuss the different data sources that Power BI supports

Data Source Type	Examples	Use Case
Files	Excel, CSV, JSON	Local or shared structured data
Databases	SQL Server, Oracle	Relational data for reporting
Online Services	Salesforce, Google Analytics	Live cloud data
Power Platform	Power BI datasets, Power Apps	Reuse datasets, automate workflows
Web / APIs	REST API, Web pages	Real-time or external data
Other / Big Data	Hadoop, Azure Data Lake	Enterprise-scale analytics

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Step 1: Open Power Query

- In Power BI Desktop, click Home → Transform Data.
- The Power Query Editor will open.

Step 2: Select the Column

- Click on the column containing full names, e.g., Owner Name.

Step 3: Split Column

- Go to Home → Split Column → By Delimiter
- Choose Space as the delimiter .
- Choose Each occurrence of the delimiter (or At the first space if you want only first & last name).

Step 4: Rename the New Columns

- The split will create two columns:
 - ✓ Owner Name.1 → rename to First Name
 - ✓ Owner Name.2 → rename to Last Name

Step 5: Apply Changes

- Click Close & Apply to save the changes and load the updated table.

Table.RenameColumns(#"Changed Type1",{{"Owner_Name.1", "Owner_First"}, {"Owner_Name.2", "Owner_Last"}})

	Area_sqft	Bedrooms	Bathrooms	Owner_First	Owner_Last	Contact
1	12500000	850	2	Rahul	Mehhta	9876543210
2	9800000	720	2	Anita	Sharma	9898989898
3	8500000	900	3	Ramesh	Kumar	9123456789
4	11000000	880	3	Neha	Verma	9988776655
5	null	950	3	Amit	Joshi	9090909090
6	7200000	800	2		null	9345678901
7	6700000	null	2	Suresh	Reddy	9012345678
8	7900000	870	3	Divya	Iyer	9556677889
9	13500000	1200	4	Manoj	Patil	9887766554
10	6500000	700	2	Raj	Singh	null
11	12500000	850	2	Rahul	Mehhta	9876543210
12	9800000	720	2	Anita	Sharma	9898989898
13	8500000	900	3	Ramesh	Kumar	9123456789
14	11000000	880	3	Neha	Verma	9988776655
15	null	950	3	Amit	Joshi	9090909090
16	7200000	800	2		null	9345678901
17	6700000	null	2	Suresh	Reddy	9012345678
18	7900000	870	3	Divya	Iyer	9556677889
19	13500000	1200	4	Manoj	Patil	9887766554
20	6500000	700	2	Raj	Singh	null
21	12500000	850	2	Rahul	Mehhta	9876543210
22	9800000	720	2	Anita	Sharma	9898989898
23	8500000	900	3	Ramesh	Kumar	9123456789
24	11000000	880	3	Neha	Verma	9988776655
25	null	950	3	Amit	Joshi	9090909090
26	7200000	800	2		null	9345678901